Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

THE EVOLUTION OF MACHINE LEARNING AND AI · WEEK 8 OF 12

Faster Language Models

Why long text is slow, what newer designs do about it, and your first model

Lecture · 2 hours · then the walkthrough video · lab at home: 1 hour

THREE PARTS, THEN YOUR LAB AT HOME

Today

1 · The topic · 1 hour

Why long text is slow, and four ideas that make language models faster.

2 · Your project · 30 min

How the topic connects, and this week’s step: your first model.

3 · The video · 30 min

A slow walkthrough in Colab, then your lab at home.

KEY IDEA

Lab at home, one hour at most. Part A (15 min): the speed visualiser. Part B (45 min): your first model. Then the Week 8 form, due before the Week 9 lecture on Monday 16 November.

LAST WEEK, AND THIS WEEK

Where we are

Last week

- A language model guesses the next token.
- Inside: a transformer, whose attention lets every word look at every other word.
- Bigger has meant better, but the walls are close: cost, energy, data.
- Your project: your law, its numbers, its start, its answer sheet.

This week

- Why attention makes long text slow.
- Four ideas for faster models.
- The lesson to keep: measure cost as well as accuracy.
- Your project: a model that learns your law with no data.

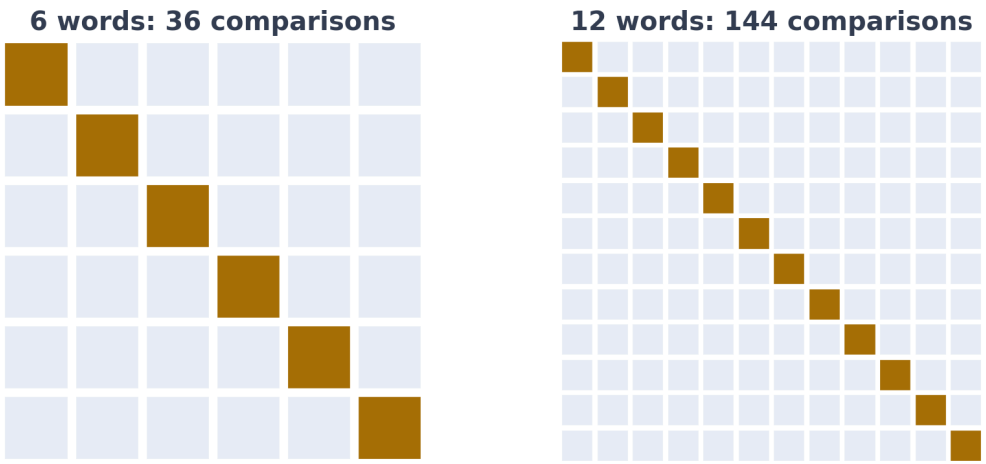
PART 1

Faster language models

Why long text is slow, and what newer designs do about it

ATTENTION: EVERY WORD LOOKS AT EVERY OTHER WORD

Why long text is slow



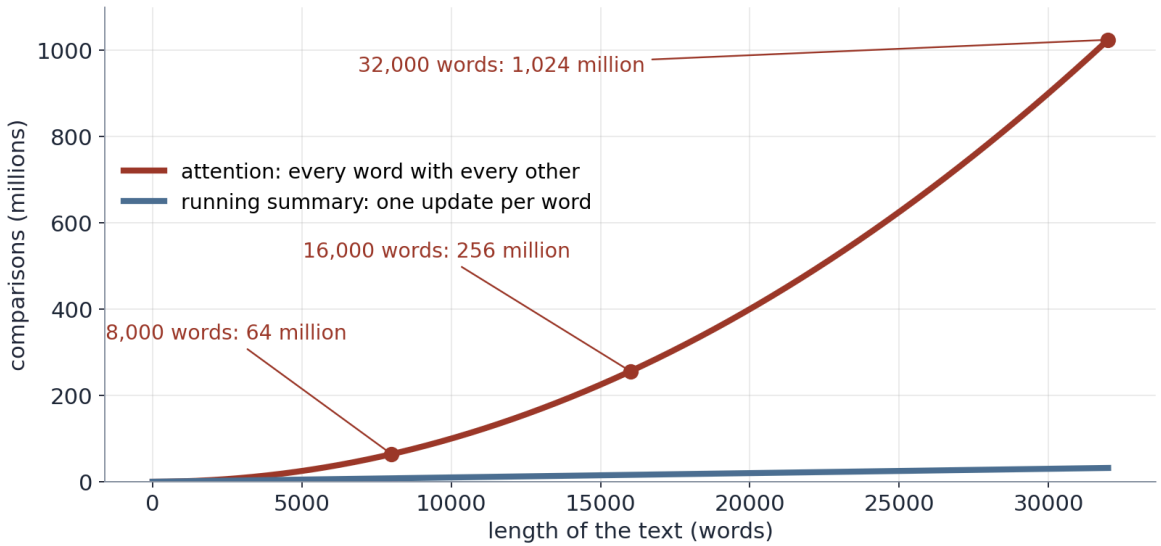
Twice the words, four times the squares to fill

KEY IDEA

Each square is one comparison: one word deciding how much another word matters to it.

IT GETS OUT OF HAND QUICKLY

Twice the text, four times the work



LONG TEXT IS WHAT PEOPLE WANT

Where this hurts

Long documents

Summarise a 300-page report, or compare two contracts.

Long conversations

A chat that remembers everything you said this term.

Large code bases

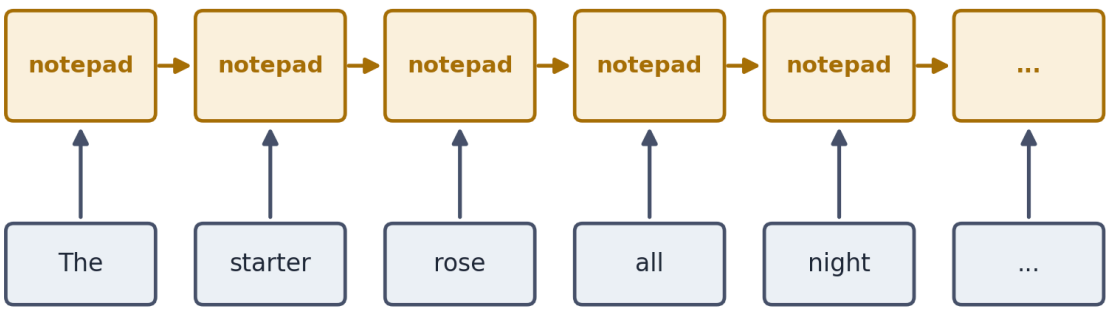
A coding assistant that reads the whole project, not one file.

The cost

More comparisons means more time, more memory, more electricity, and a bigger bill per question.

Idea 1: read with a notepad

Read one word, update a small notepad, move on. The notepad never gets bigger.

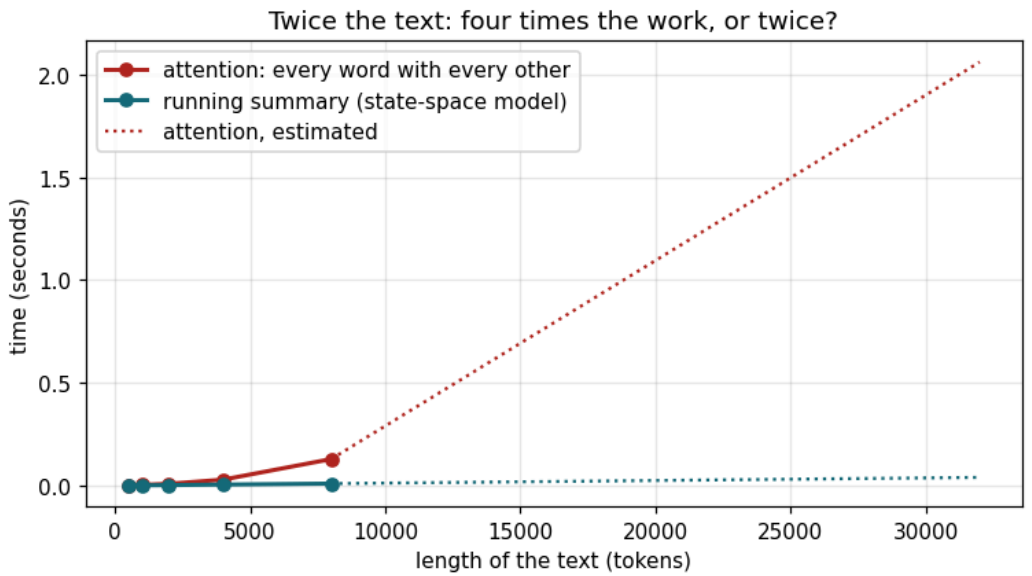


EVERYDAY ANALOGY

Like reading a novel and keeping a page of notes as you go, instead of re-reading every earlier page for every new sentence.

MEASURED ON A LAPTOP: YOU WILL RUN THIS IN PART A

Twice the text, twice the work



EVERY DESIGN IS A TRADE-OFF

What the notepad forgets

Attention

Can look back at any word, exactly. Expensive for long text.

Notepad

Cheap for long text. But a small notepad cannot hold every detail.

KEY IDEA

Hybrids use both: mostly notepad layers, with a few attention layers to look back exactly when needed. Jamba (2024) is one example.

COST THAT GROWS ONLY IN STEP WITH THE TEXT

Other designs with the same aim

RetNet (2023)

A transformer-like model that can also run as a reader with a memory.

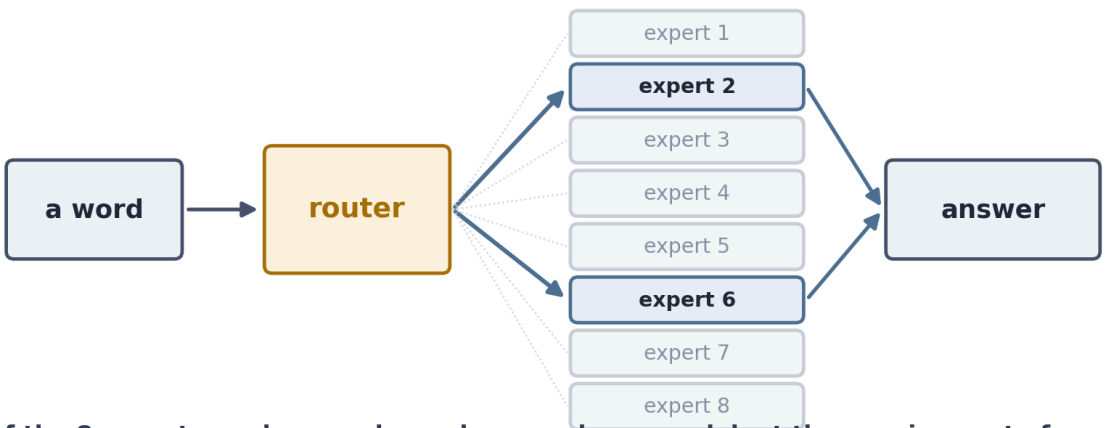
RWKV

An open, community-built model that reads in order, like a recurrent network, but trains like a transformer.

THE LAW

The shared idea: **twice the text should cost about twice as much, not four times**.

Idea 2: a team of experts

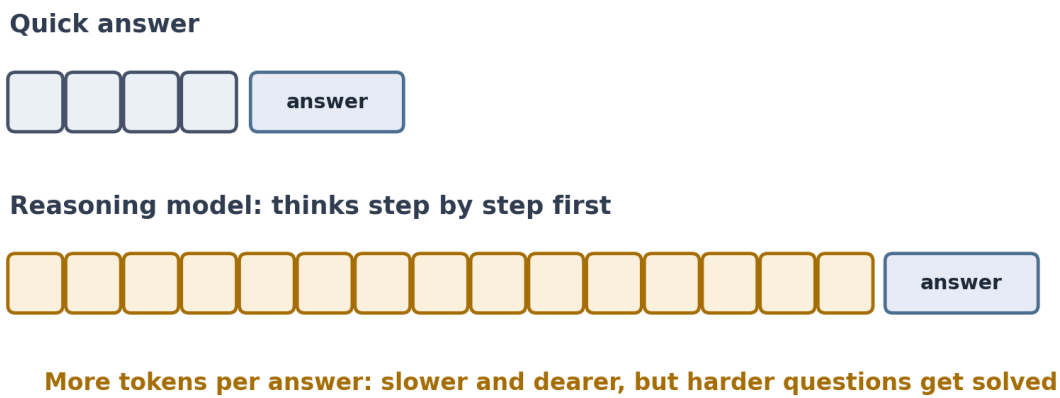


Only 2 of the 8 experts work on each word: a very large model, at the running cost of a small one

KEY IDEA

Example: Mixtral 8×7B has about 47 billion numbers in total, but uses only about 13 billion for each word.

Idea 3: think for longer



KEY IDEA

Examples: OpenAI o1 (2024) and DeepSeek-R1 (2025) write out their steps before answering.

WHAT EACH SAVES, AND WHAT IT COSTS

Four ideas, side by side

Idea	Saves	Costs
Notepad (state-space)	Time and memory on long text	Can forget exact details
Other attention-free designs	Time and memory on long text	Newer, less tested
Mixture of experts	Computing time per word	Memory: all experts stored
Reasoning models	Wrong answers to hard questions	Time and money per answer

ALWAYS MEASURE COST AS WELL AS ACCURACY

The lesson to keep

Accuracy

- error against the answer sheet
- success test: passed?

Cost

- training time
- model size (how many numbers)
- memory

★ KEY POINTS TO REMEMBER

▪ A model that is a little less accurate but ten times cheaper can be the better choice. You can only know if you measure both.

FIVE THINGS TO REMEMBER

Part 1, done

1. Attention compares every word with every other: **twice the text, four times the work**.
2. State-space models read with a small notepad: **twice the text, twice the work**. Hybrids mix both.
3. Mixture of experts: a big model that runs only a few experts per word.
4. Reasoning models think longer: dearer per answer, better on hard questions.
5. **Measure cost (time, size, memory) as well as accuracy.**

PART 2

How it connects to your project

Your first model, and its cost

FROM THIS WEEK, RECORD COST AS WELL AS ACCURACY

The same lesson, in your project

Accuracy

Your error against the answer sheet, and whether your success test passes.

Cost

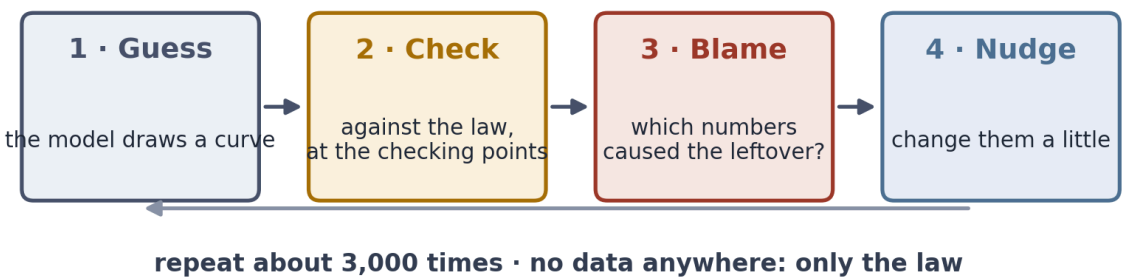
Your training time, and your model’s size: how many numbers it has.

KEY IDEA

Week 9 needs both: you will race your model against one that learns from examples, on accuracy, speed and size.

IT LEARNS YOUR LAW, WITH NO DATA

This week’s step: your first model

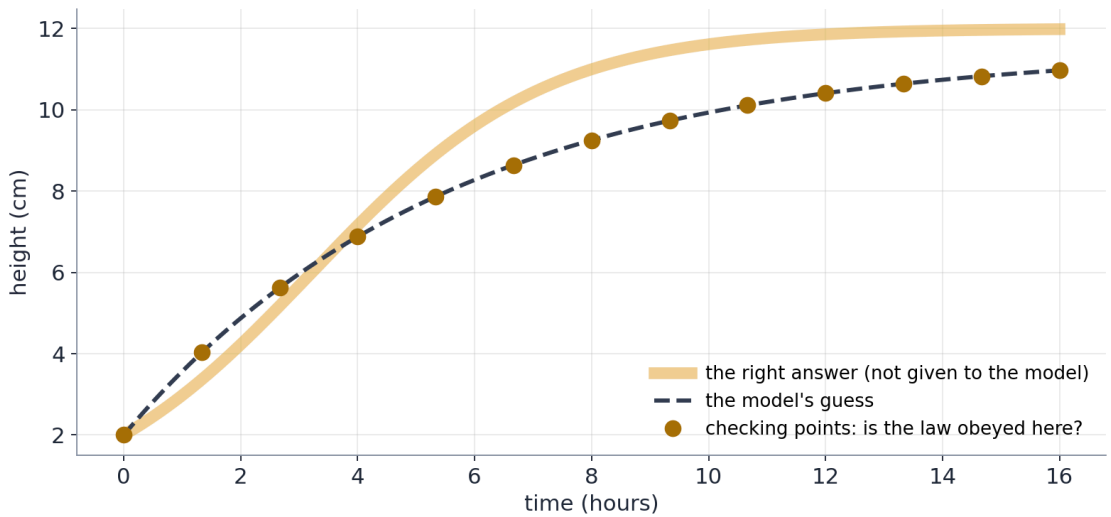


THE LAW

Exactly as in Week 5: the only teacher is the law.

THE LAW IS CHECKED AT MANY MOMENTS IN TIME

Checking points and the leftover



KEY IDEA

The guess (dashed) is checked at each orange dot. Training shrinks the leftover until the guess obeys the law everywhere.

THE SAME THREE SHAPES AS LAST WEEK

Built-in start, and the model’s shape

One thing changes

One input (time), one output. Part 2 of the notebook.

Two or three things

One input (time), one output per thing. Part 3.

Along a line

Two inputs (place and time), one output. Part 4.

KEY IDEA

The start is built in (the fade-in): the model always begins exactly at your starting value, so it only has to learn what happens next.

A NUMBER, SET BEFORE YOU TRAIN

Your success test, decided first

The class example The height at **8 hours** is within **0.1 cm** of the answer sheet.

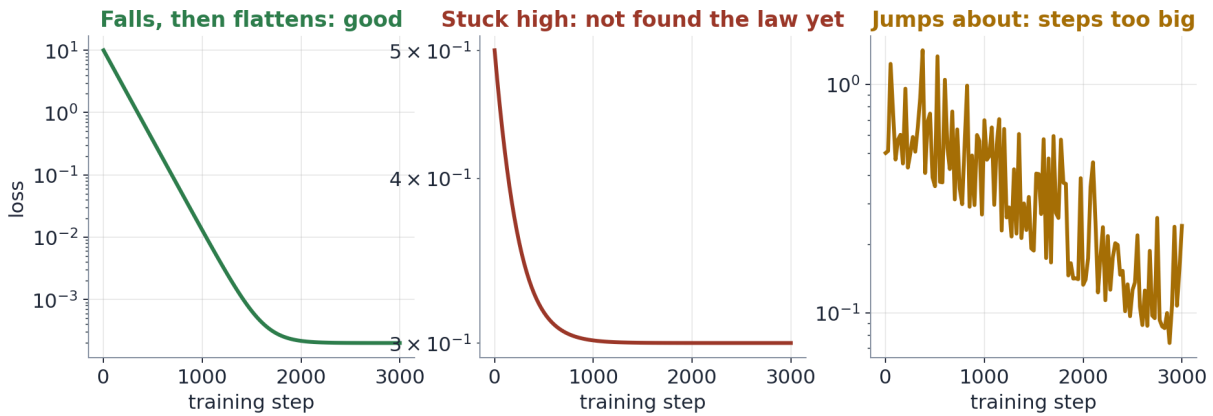
Yours: the same shape At **time …**, the model is within **… units** of the answer sheet. Your roadmap suggests one.

Why first?

So you cannot move the goalposts after you see the result.

THREE SHAPES, THREE MEANINGS

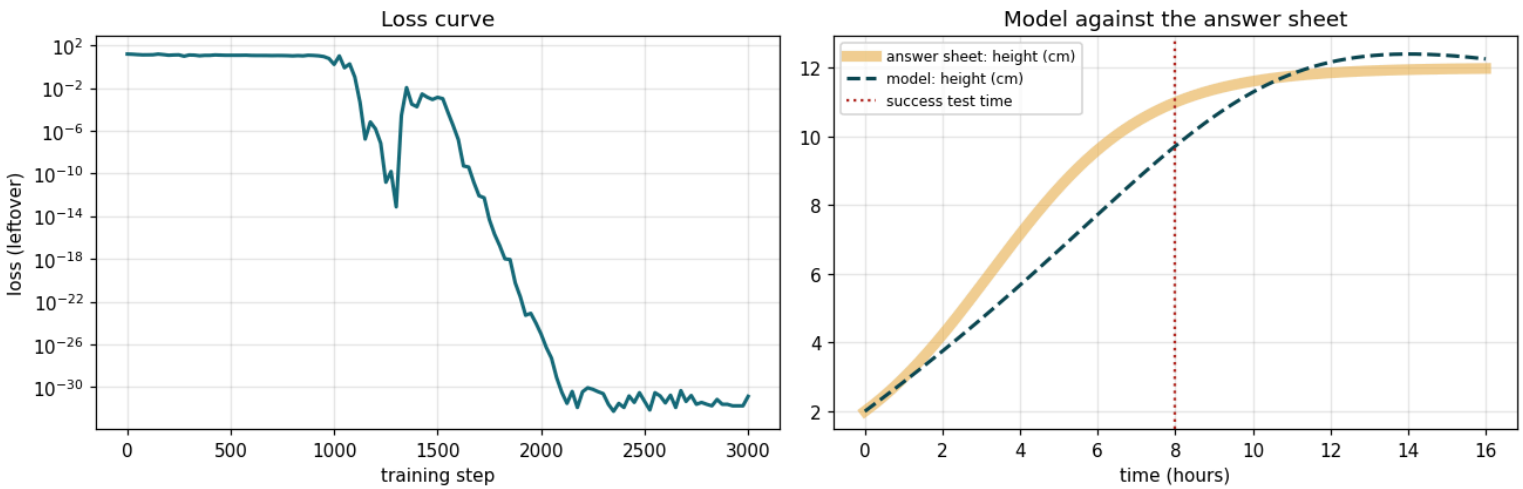
Reading the loss curve



KEY IDEA

The loss is the average leftover. In the real class example it is bumpy at first, because the model learns the early hours first. What matters is the end: it falls, then flattens.

Break it on purpose: 3 checking points



What happened?

The law was obeyed perfectly at 3 points, and nowhere else. The success test failed: 1.27 cm off, where 0.1 is allowed.

TRY ONE AT A TIME

If it fails: four fixes, in order

1. **Keep the numbers near 1.** Choose sensible units: hours, not seconds.
2. **Use more checking points.** The default is 200.
3. **Use a shorter time range.** Look at less time first.
4. **Take smaller steps.** A smaller learning rate, for example `lr=0.003`.

> **KEY IDEA**
>
> Fixed: with 200 checking points, the class example passes: 0.0007 cm off at 8 hours. Note the one problem you fixed: the form asks for it.

TWO OR THREE SENTENCES, ON THE FORM

Applied checkpoint: where could it go wrong?

What the law leaves out

Your model cannot know it. Say it.

The numbers you chose

Where they came from, and how sure you are.

Week 5's lesson

A model trained on a wrong law is confidently wrong.

KEY IDEA

This short report is formative feedback, and it becomes part of section 6 of your final report.

PART 3

The walkthrough video

Your first model in Colab, step by step, then your lab at home

THE VIDEO GOES THROUGH THESE SLOWLY; IT IS ON TUTORS TO REWATCH

The walkthrough, in six steps

1. Open `capstone_step2.ipynb` in Google Colab, and save your own copy.
2. Run Step 0 (the tools), and Part A (the speed visualiser).
3. Run the class example: train it, read the loss curve, check the success test.
4. Break it on purpose (3 checking points), then fix it.
5. In your part (2, 3 or 4), copy your Week 7 lines and set your success test.
6. Train your first model, and download `my_first_model.png`.

ONE HOUR AT MOST · DUE MONDAY 16 NOVEMBER, BEFORE THE WEEK 9 LECTURE

Your lab at home

Part A · 15 min · the topic

Run the speed visualiser. Which is faster for long text, and why?

Part B · 45 min · your project

Your first model: train it, check your success test, download the picture.

KEY IDEA

Then the Week 8 form on Tutors.

WHAT TO HAND IN

The Week 8 form

- Part A: which design is faster for long text, and why, in a sentence.
- Your **error**, and your **success test**: passed, or not yet.
- Your **training time** and **model size**.
- The picture `my_first_model.png`.
- **One problem you fixed**, and how.
- **Where your model could go wrong**: two or three sentences.

This week, in five lines

1. Attention: twice the text, four times the work.
2. Notepad models (state-space): twice the text, twice the work; hybrids mix both.
3. Mixture of experts runs a few experts per word; reasoning models think longer.
4. Measure cost as well as accuracy.
5. Your first model: it learns your law with no data, and passes (or not yet) your success test.

This week, train your first model.

Next week: smaller language models, and the race against a model that learns from examples.

Questions any time in the channel.